Client Package — Contents

Tosco Branch Re-Entry Wells — Production Potential, Geology & Drilling Execution

Bowman County, North Dakota · Williston Basin southern margin (Red River trend)

Prepared for: Marlo Operating Company · Valor Energy Partners | Wells: Branch 4 RR (#41700) · Branch 2 (#41258)

HOW TO READ THIS PACKAGE

Start with the Executive Summary (1) for the bottom line. Documents 2–3 answer “will it produce and with which completion”; document 4 answers “where to drill”; documents 5–7 answer “how the well will drill” — mechanics, fluid balance, and steering. Scope is the probability of a producing well and how to execute it — not economics.

Executive Summary

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Bottom line

Both wells are RE-ENTRIES into a pool Marlo just discovered and is already producing (sister well Branch 5). That makes them DEVELOPMENT-CLASS step-outs, not wildcats: the reservoir is effectively proven, and the outcome turns on EXECUTION of the short-radius re-entry rather than on whether oil is present. We assess a ~64–72% probability of a successful producer (modest size), with the dominant risk being mechanical — and largely knowable in advance from the original wellbore's logs.

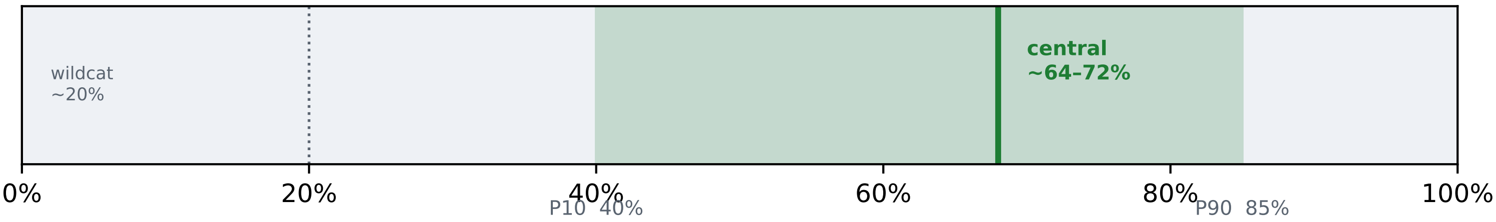
~64–72%
Probability of a successful producer

Development-class re-entry
Not a wildcat (pool is producing)

Execution
Dominant risk: short-radius mechanics

~65–150 bopd
IP if successful · EUR 40–150 Mbbl (offset Branch 5 ≈65; discovery ~497 bopd)

Chance of success — development-class, well above the ~20% wildcat line



RECOMMENDED ACTION

Pull the original Branch 4 wellbore's casing / cement-bond log and confirm the target zone. That one step sets the dominant (mechanical) risk and collapses most of the 40–85% band toward the high end before committing the curve.

EXECUTION RISK — TRACKED IN THREE LAYERS

Mechanical
Front-loaded — kickoff & casing integrity, knowable in advance from the original wellbore logs.

Geological
Landing and holding the thin 'B'; the offset / fleet dip prior keeps the lateral in zone.

Fluid — loss & gain
Mud lost to natural fractures; water gain at the OWC — held with MPD / LCM / ECD discipline.

IN THIS PACKAGE — SEVEN DOCUMENTS

full index & page references on the Contents page

1. Executive Summary (this page)
2. Production-Probability & Execution Report
3. Production Expectation Matrix
4. Geologic / Geosteering Window Report
5. Drilling Execution Forecast
6. Fluid Balance — Loss & Gain Risk
7. Steering Anticipation at Scale

Tosco Branch Re-Entry Wells

Production Potential Based on Execution

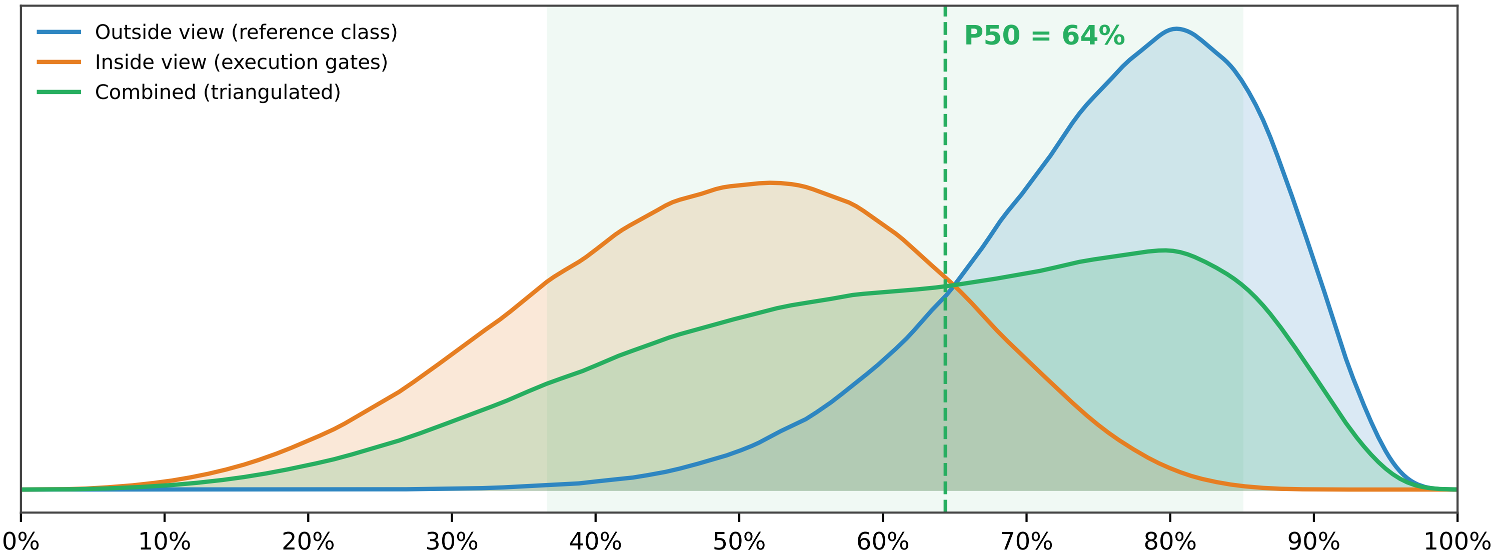
Bowman County, North Dakota • Williston Basin southern margin (Red River trend)

BRANCH 4 RR	BRANCH 2
NDIC #41700 • API 33-011-01562	NDIC #41258 • permitted 06/11/2026
Status: DRILLING (06/19/2026)	Status: PERMITTED (re-entry)
Re-entry • short-radius (25-30 deg/100ft)	Re-entry into same Marlo pool
Sec 19-129N-103W	BHL Sec 20-129N-104W

Bottom line

Both wells are RE-ENTRIES into a pool that Marlo just discovered and is already producing (sister well Branch 5, NDIC #41701). That reclassifies them from 'wildcat' (~20% success) to DEVELOPMENT-CLASS step-outs. The reservoir is effectively de-risked; the bet is EXECUTION of the short-radius re-entry. Same intrinsic potential for both wells — Branch 4 is further along (drilling now), Branch 2 is at the start (permitted), so Branch 4's risk is retiring first.

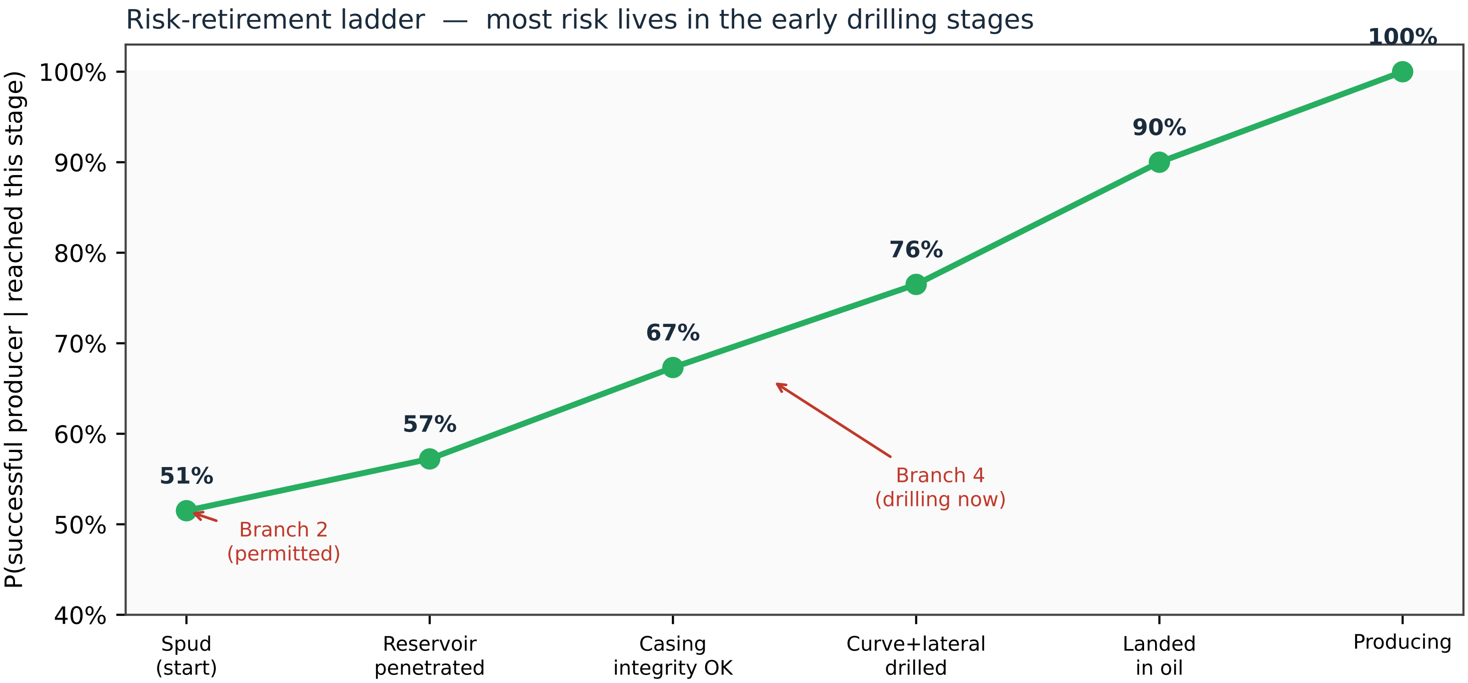
Chance of success — three independent estimates



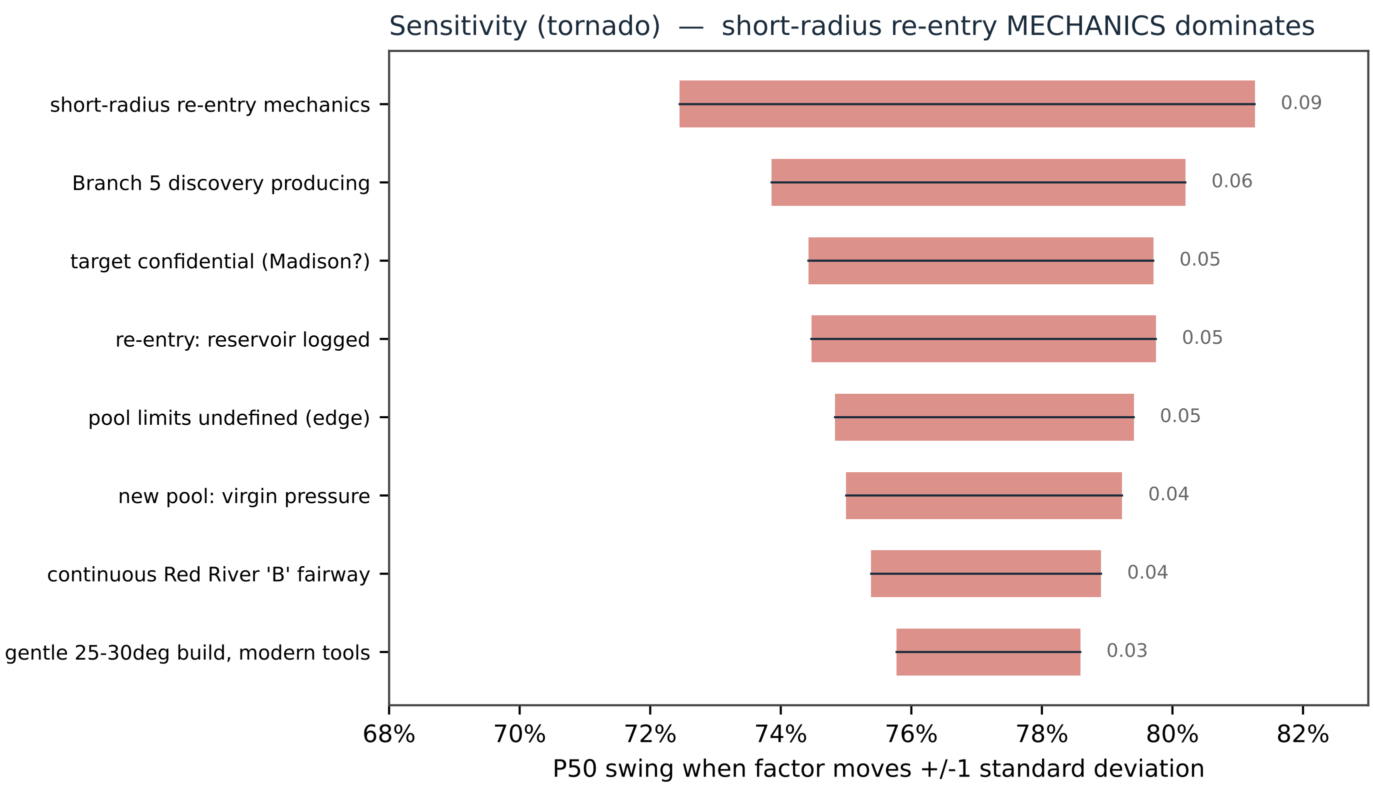
Probability of a successful producing well (combined P50 64%; 80% band 37%-85%)

Dynamics of Execution — how risk retires

Probability climbs as each well passes its drilling and completion gates.



Read: the biggest lever is mechanical execution of the re-entry — the risk the original wellbore's casing/cement-bond log can resolve before the curve is ever drilled.



The Mathematics, Explained

A transparent two-estimator Bayesian Monte-Carlo. Inputs are structured judgment in log-odds; the reference class is empirical.

$$p_0 \sim \mathrm{Beta}(s+1,\ f+1)$$

Reference-class prior. Start from how often analog wells actually succeeded: s successes, f failures (NDGS Red River development/extension = 20 of 27). The empirical anchor.

$$p = \sigma\Big(\mathrm{logit}(p_0) + \sum_i \Delta_i\Big), \quad \sigma(x) = \frac{1}{1+e^{-x}}$$

Outside view. Combine evidence in LOG-ODDS (logit): each factor delta_i nudges the odds up (re-entry, adjacent producer) or down (re-entry mechanics, confidential target). sigma maps back to a probability.

$$P_{\mathrm{in}} = \prod_{k=1}^{5} g_k, \quad g_k = \sigma\Big(\mathrm{logit}(b_k) + \lambda_k^e q_e + \lambda_k^g q_g + \varepsilon_k\Big)$$

Inside view. A chain of 5 execution gates (reservoir -> casing -> curve -> placement -> rate). Two SHARED latent factors q_e (execution quality) and q_g (geology) make the gates rise and fall together (correlation).

$$\mathbb{E}[\sigma(\mathrm{logit}(b) + \varepsilon)] < b \quad (b > \tfrac{1}{2})$$

Jensen's inequality. The logistic is concave above 0.5, so UNCERTAINTY drags each gate's expected pass-rate BELOW its nominal value. This is why the simulated inside-view mean (0.49) sits just under the naive product (0.515).

$$\{p\} = \{p_{\mathrm{out}}\} \cup \{p_{\mathrm{in}}\} \ \Rightarrow\ P_{10},\, P_{50},\, P_{90}$$

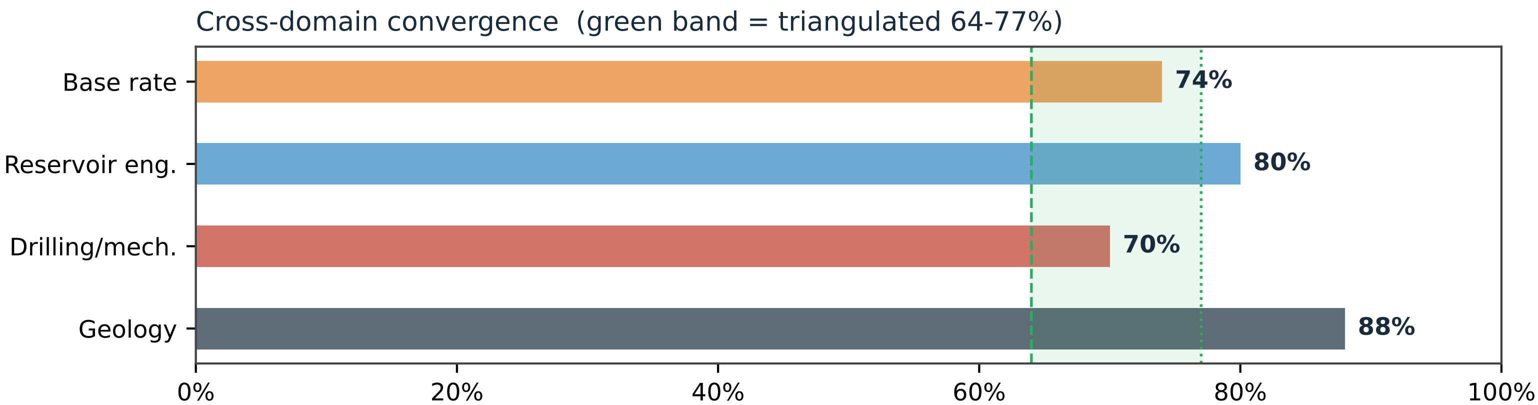
Triangulation. Pool the two independent sample sets and report the distribution. Their divergence (0.77 vs 0.50) is diagnostic: it localizes the uncertainty to execution.

$$P(\mathrm{success} \,|\, \mathrm{gate}\ k\ \mathrm{passed}) = \prod_{j>k} b_j$$

Dynamic update (Bayes). Once a gate is observed to pass, drop its risk; the conditional probability climbs (the ladder on p.2: 51% -> 57% -> 67% -> 77% -> 90%).

Verification, Validation & Sources

Four independent domains converge; verified (no leakage) and validated against field analogs.



Verified: all model numbers reconcile with the Monte-Carlo output; forward-generative (no fitted predictor, no data leakage); inside/outside divergence reported, not averaged away. Independently re-checked by a fresh-context verifier (one propagated wording error on the Jensen effect was found and corrected). Validated: the DOE/Luff Bowman re-entry program (0 of 4 dry from no-reservoir, ~1 of 4 clearly economic) corroborates BOTH the high reservoir CoS and the heavy mechanical/placement discount.

Selected citations

[1] NDIC Oil & Gas Division — Daily Activity Report 06/19/2026 (File #41700, re-entry, DRILLING); Hearing Docket 04/22/2026, Case #32729 (Branch 5 discovery, field-limit spacing). dmr.nd.gov/oilgas

[2] N. Dakota Geological Survey, 'Oil & Gas Potential of the Red River Fm, SW North Dakota' (2017) — development/extension success 70-86% (Camel Hump 6/7; step-out 14/20).

[3] Carrell, George & Gibbons (1997), DOE / Luff Exploration, 'Lateral Drilling & Completion Tech., Red River & Ratcliffe, Williston Basin', OSTI #2214 — short-radius re-entry case results & costs.

[4] Oil & Gas Journal, 'Horizontal projects buoy Williston recovery' — Cedar Hills Red River 'B' (IP, EUR, RF).

[5] AAPG Datapages — 'The Red River B Zone at Cedar Hills Field, Bowman County, ND' (reservoir character).

[6] USGS — 'Geology & Undiscovered Oil & Gas Resources, Madison Group, Williston Basin' (Mission Canyon).

[7] DrillingEdge / ShaleXP — Bowman County & Marlo Operating activity (Branch 5 producing, Apr 2026).

[8] Reference-class forecasting / 'outside view' — Kahneman & Tversky (1979); Flyvbjerg (2006).

Scope: probability of a producing well, NOT economics. Inputs are expert-judgment priors except the empirical reference class. Target formation confidential to 12/19/2026; assessment assumes the Red River trend. Magnitude if successful: IP ~tens-to-low-hundreds bopd (offset Branch 5 ≈ 65 bopd), EUR ~40–150 Mbbl.

Production Expectation Matrix

By well & completion strategy — anchored on the execution-stage chance-of-success (same breakdown as the Production-Probability Report)

Scenario	P(success)*	Expected production	Geologic fit	Dominant risk	Verdict
Conventional re-entry open-hole / acid (Red River 'B')	64–72%	40–150 Mbbl · IP tens–low-100s bopd flat 5–15%/yr decline, long life	EXCELLENT thin fractured dolomite	short-radius re-entry mechanics (casing / curve)	RECOMMEND
New-drill horizontal acid (Red River 'B')	70–78%	80–200 Mbbl cleaner lateral placement	Excellent	higher cost; loses the re-entry savings	STRONG ALT
Hydraulic frac multi-stage proppant (RR 'B')	45–60%	high variance; water-cut risk dominates	POOR thin carbonate over OWC	frac propagates DOWN into the water leg	AVOID
Madison / Mission Canyon conventional target	50–62%	facies-dependent (porosity-cycle controlled)	Moderate trap/porosity-controlled, patchy	in / out of the porosity fairway	CONDITIONAL
Bakken / Three Forks frac (basin margin)	25–40%	sub-core EURs on the fairway edge	Marginal edge of the Bakken fairway	play immature here; high capex	AVOID HERE
Vertical re-entry no lateral	70–80%	low (8–15% recovery factor) thin pay	n/a	low rate / sub-commercial	LAST RESORT

* P(success) = execution-stage chance of success (reservoir → casing → curve → lateral placement → rate). For conventional scenarios the dominant lever is short-radius re-entry execution, NOT reservoir preser

GEOLOGY OF FRACKING HERE — why it is the wrong tool for the 'B'

The Red River 'B' is a thin (~6–10 ft) NATURALLY-FRACTURED dolomite sitting directly above tight limestone and the oil-water contact. It produces through its natural fractures + acid — not proppant. A multi-stage hydraulic frac in a carbonate this thin and this close to the OWC propagates fractures DOWN into the water leg: you trade oil for water cut, and add mechanical and cost risk for little upside. Hydraulic fracking is essential in the tight Bakken / Three Forks shale — but Bowman County is on the MARGIN of that fairway, where Bakken economics are sub-core. Right tool here: lateral exposure + acid, geosteered to stay in the upper 'B' above the OWC (see the Geologic Window report).

READING THE MATRIX

Green = recommended completion for this rock.

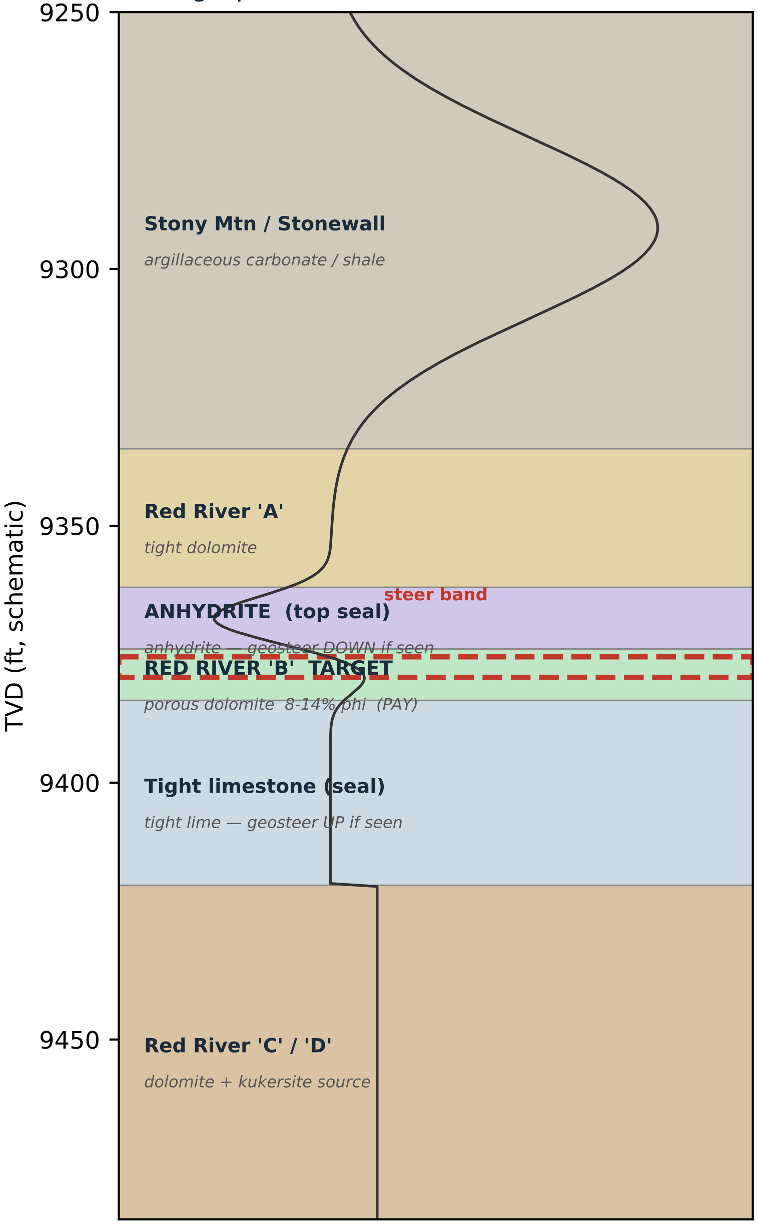
Amber = conditional — only with seismic / log support.

Red = wrong tool or wrong place (fracking the 'B'; Bakken on the margin).

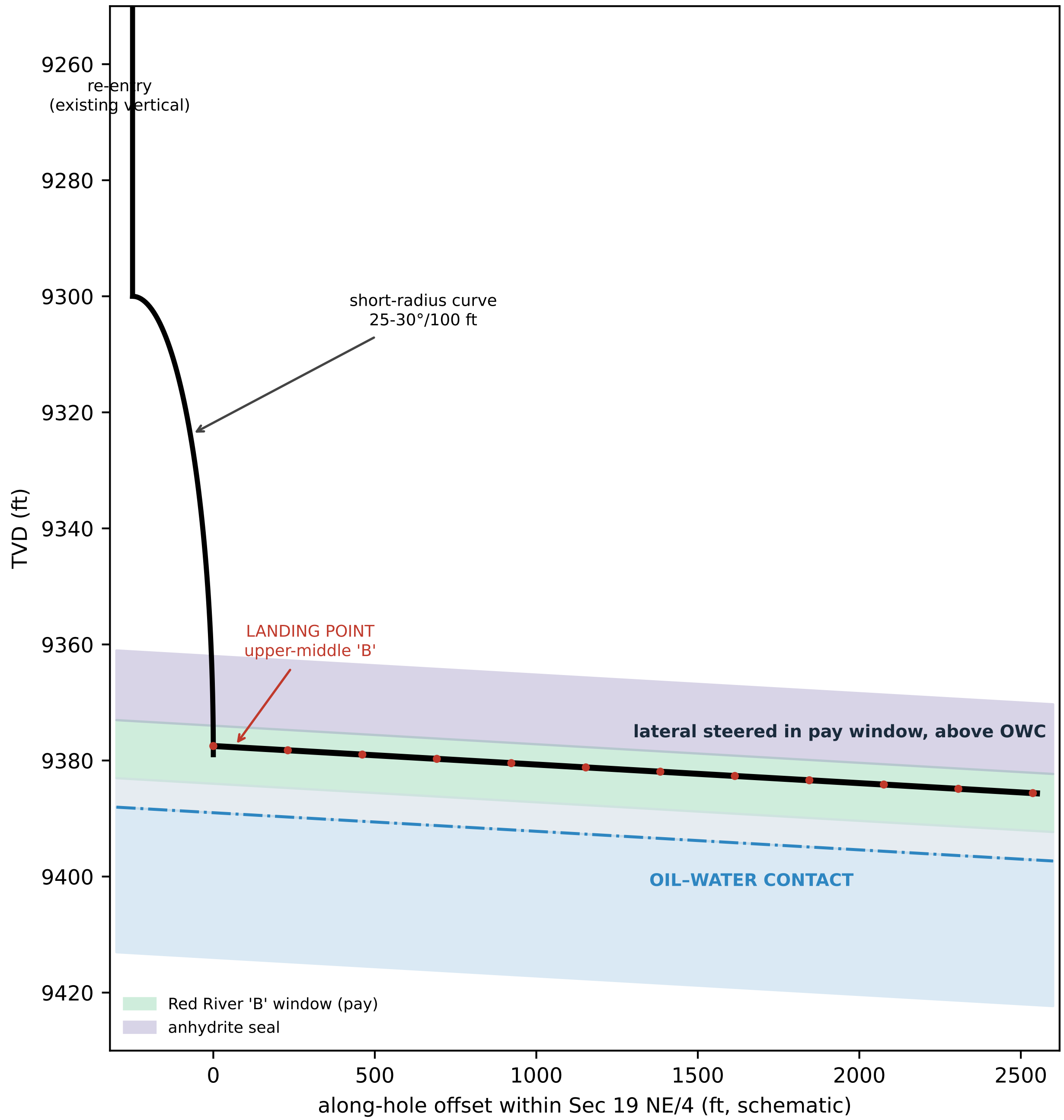
Tosco Branch 4 RR — Geologic Window for Drilling

Keep the lateral inside the Red River 'B' porous dolomite — between the anhydrite seal above and the tight lime / OWC below — for maximum oil contact.

Stratigraphic column + schematic GR



Geosteering cross-section — stay in the 'B' window



Geosteering Playbook — staying in window for best effect

Marker signatures, steering rules, and the optimization that maximizes oil-saturated reservoir contact.

Marker beds & log signatures (what the MWD gamma is telling you)

Marker	Log signature	Steering action
Anhydrite SEAL (above 'B')	very low GR, high density/PE, no porosity	you are HIGH — steer DOWN into the 'B'
Red River 'B' PAY (target)	low-moderate GR, neutron-density porosity separation (dolomite)	ON TARGET — hold the steer band
Tight limestone (below 'B')	low GR, no porosity, density tight	you are LOW — steer UP; OWC risk below
Oil–water contact (OWC)	resistivity drop / increasing water on shows	STOP descending — stay updip / above

Steering rules — the window

- Target the UPPER-MIDDLE of the 'B' (≈2-6 ft below the anhydrite seal): maximum standoff from the OWC while keeping a buffer below the seal.
- Maintain steer band within the ±2-3 ft 'sweet spot'; the 'B' is only ~8-10 ft thick, so small dip changes move you out of zone.
- FOLLOW STRUCTURAL DIP using the MWD gamma + correlation to the original wellbore and Branch 5 tops — adjust inclination to track the dipping 'B'.
- Never drill below the OWC: descending into the water leg trades oil footage for water cut (the #1 re-entry failure mode in this play).

Best effect for the client

Maximize productive lateral footage inside the porous, oil-saturated 'B' dolomite. Every foot held in the steer band is reservoir contact; every foot in the seal or water leg is wasted hole. A clean in-zone lateral in the upper 'B', updip of the OWC, delivers the field's best per-well recovery (Cedar Hills 'B' analog: low initial water, 5-15%/yr decline, long life). The geologic upside is realized in the STEERING, not just the target pick.

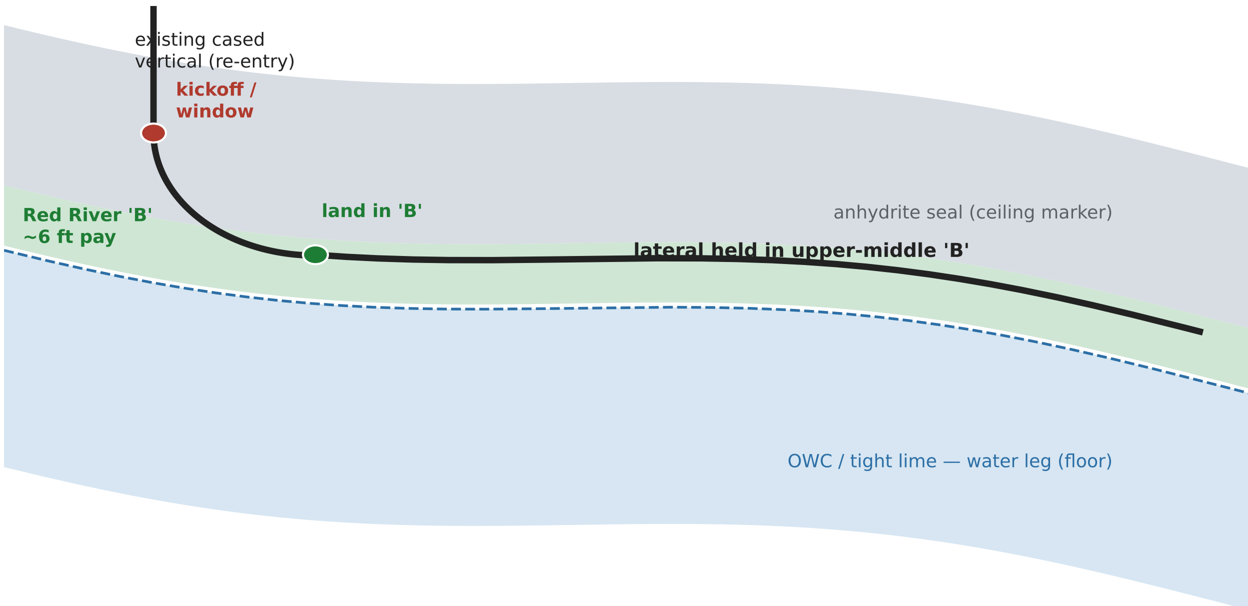
Calibration required (honest scope)

Depths, tops and the OWC above are a PLAY-MODEL SCHEMATIC (public Cedar Hills Red River 'B' character + this well's parameters). Before drilling, tie to: (1) the original Branch 4 wellbore logs (actual 'B' top, thickness, porosity);
(2) Branch 5 tops + fluid contacts (the OWC datum); (3) a structural dip map. Target zone confidential to 12/19/2026.

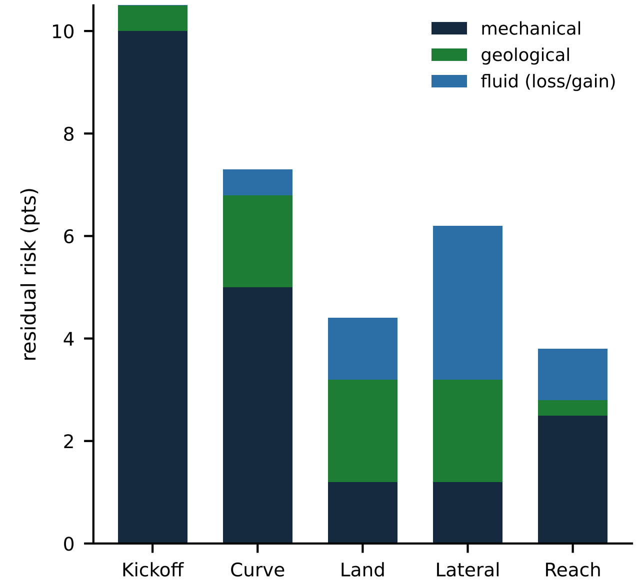
Drilling Execution Forecast

How the short-radius re-entry should drill — geometry × geology, section by section, and where the residual risk sits

Geometry through the geology: vertical re-entry → short-radius curve → thin-pay lateral



Where the ~32% residual risk sits
(Σ ≈ 32% → P(success) ~ 68%)



1 · Re-entry & Kickoff

- **GEOM** cut window in casing; sidetrack
- **GEOL** steel & cement, not rock
- **ROP** n/a
- **RISK** casing / cement-bond integrity

2 · Short-radius Curve

- **GEOM** high build rate / DLS; tool-face fight
- **GEOL** abrasive anhydrite ↔ dolomite cycles
- **ROP** slow; real bit wear
- **RISK** exit curve off-TVD; torque & drag

3 · Land in the 'B'

- **GEOM** flatten into a ~6-ft window
- **GEOL** anhydrite ceiling, OWC floor
- **ROP** —
- **RISK** land high (no pay) / low (water)

4 · Lateral in Pay

- **GEOM** hold upper-middle 'B'
- **GEOL** fractured dolomite — drills fast
- **ROP** fast
- **RISK** lost circulation; thin-zone steer

5 · Reach / Hole Mechanics

- **GEOM** T&D limits lateral length
- **GEOL** easy cleaning at angle
- **ROP** steady
- **RISK** torque/drag; ECD vs losses & OWC

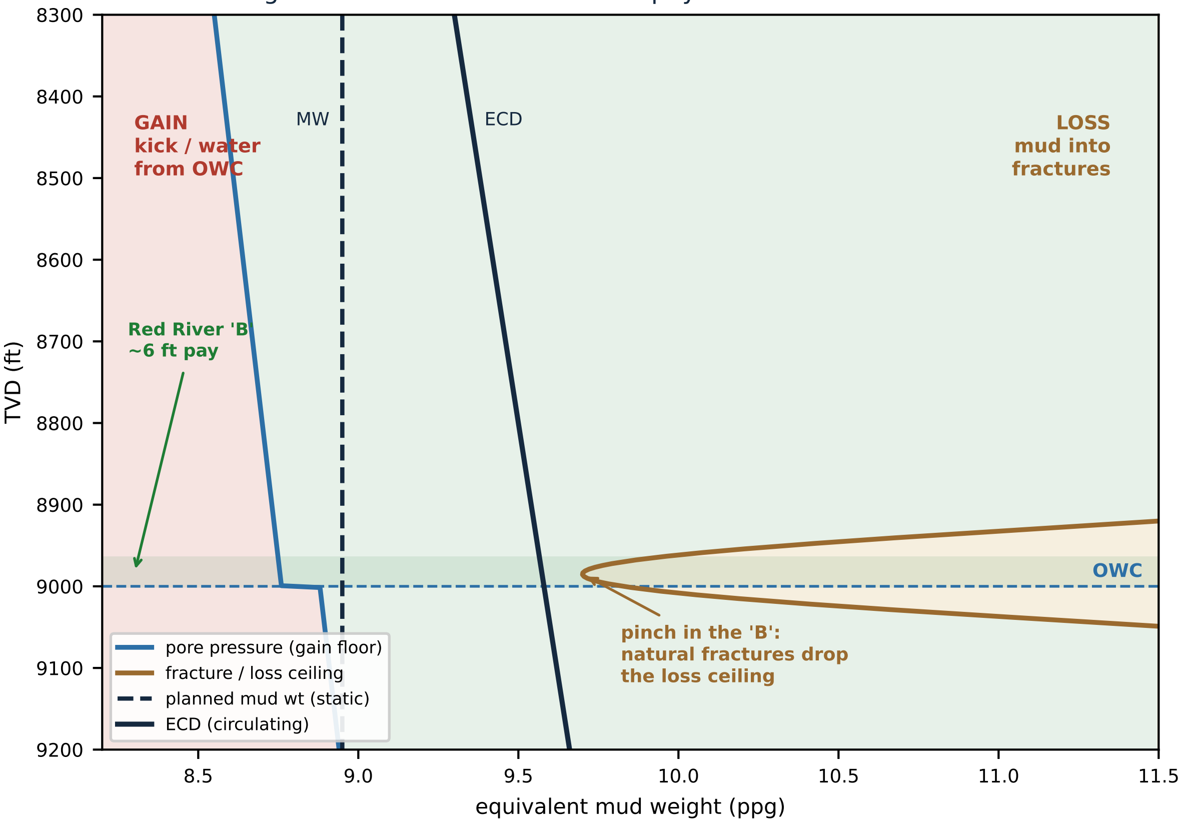
WHAT RETIRES THE RISK — before the curve is cut

- ✓ **Original Branch 4 CBL / casing log**
  retires kickoff & casing risk — the biggest single lever
- ✓ **Pilot-hole logs ('B' / anhydrite / OWC TVD)**
  retires landing risk — land to a known depth
- ✓ **Offset / fleet dip model (sheaf prior)**
  retires steering lag — holds ~95% in-zone

Fluid Balance — Loss & Gain Risk

Drilling-system fluid (mud / ECD) vs formation pressure — the mud-weight window pinches inside the pay, where losses and gains meet

The mud-weight window narrows inside the pay



TWO WAYS THE FLUID BALANCE FAILS

OVERBALANCED — ECD above formation pressure

→ LOSSES: mud invades the natural fractures; you lose returns, burn mud, and can drill blind to a kick building below.

UNDERBALANCED — ECD below formation pressure

← GAIN / KICK: formation fluid enters the well; here that means WATER pulled up from the OWC sitting just under the 'B'.

LOSS / GAIN RISK BY PHASE

phase	LOSS	GAIN
Curve	Low	Low
Landing	Med	Med
Lateral	High	Med
Trips / connections	Med	Med

LOSS MANAGEMENT

- LCM sized to the fractures, staged & ready
- hold ECD down: flow rate, hole cleaning, RPM, slow connections & trips (surge)
- cure losses before drilling ahead — don't chase
- air-mist / UBD only where clear of the OWC

GAIN / WELL CONTROL

- flow checks + active kick detection
- trip & connection discipline (avoid swab)
- keep a trip margin; PWD to watch BHP live
- geosteer to hold standoff ABOVE the OWC — the cheapest gain insurance there is

NARROW-WINDOW TOOL · MPD

Managed-pressure drilling: closed-loop, holds a precise bottomhole pressure and reacts in seconds to BOTH losses and gains — the right tool where the loss ceiling and gain floor pinch in the pay. With the OWC right below, hold a precise slight overbalance.

Extended Steering Anticipation — at Scale

Anticipation only pays AT SCALE: steering on a noisy single-well forecast backfires; a fleet-learned dip prior is what makes it win.

Finding: anticipation pays ONLY through a reliable, fleet-learned dip prior (the sheaf/structural model, this repo): in-zone climbs 71%→98% as the prior strengthens, and the fleet mean beats reactive (90%→95%). Anticipating off single-well data is WORSE than reacting; extending the look-ahead horizon does NOT help (best ~0 ft).